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#ifndef ARTERY_TRAJEKTORIAPPVERIND_H_
#define ARTERY_TRAJEKTORIAPPVERIND_H_

#include "artery/application/ItsG5BaseService.h"
#include "artery/application/VehicleDataProvider.h"
#include "KalmanFilterIND.h" // <- PENTING: Include file KF kita
#include <omnetpp/simtime.h>
#include <deque>
#include <fstream>
#include <string>
#include <map>
#include <vector>
#include <memory> // Untuk std::unique_ptr

namespace artery
{

// Struct data mentah (Sama seperti v1)
struct MovementData {
    omnetpp::simtime_t timestamp;
    double latitude;
    double longitude;
    double speed_mps;
};

// Struct history (Dimodifikasi untuk KF)
struct AgentHistory {
    std::deque<MovementData> history; // Masih berguna untuk log
    mentah
    omnetpp::simtime_t lastReceptionTime;
    bool hasNewData = false;

    // --- PERUBAHAN UTAMA ---
    // Setiap agen sekarang memiliki instance Kalman Filter-nya
    sendiri
    std::unique_ptr<KalmanFilterIND> kf_state;
    // -----
};

// Ganti nama class
class TrajektoriAppVerIND : public ItsG5BaseService
{
public:
    void initialize() override;
    void finish() override;

protected:

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        void handleMessage(omnetpp::cMessage* msg) override;
        void receiveSignal(omnetpp::cComponent* source,
omnetpp::simsignal_t signalID, omnetpp::cObject* obj,
omnetpp::cObject* details) override;

private:
    // Fungsi lama untuk log mentah (tidak berubah)
    void logTrajectory();
    std::string getNodeType();

    // --- PERUBAHAN UTAMA ---
    // Ganti nama dari logCoefficients menjadi runPredictions
    void runPredictions();
    // -----

    // Variabel lama (tidak berubah)
    omnetpp::cMessage* mLogTimer = nullptr;
    omnetpp::simsignal_t mCamReceivedSignal;
    omnetpp::SimTime mLogInterval;
    std::ofstream mLogFile;
    std::map<long, AgentHistory> mOtherNodes;
    const artery::VehicleDataProvider* mVehicleDataProvider =
nullptr;

    // --- PERUBAHAN UTAMA ---
    // Ganti nama timer dan file log
    omnetpp::cMessage* mPredictionTimer = nullptr;
    std::ofstream mPredictionLogFile;
    // -----
};

} // namespace artery

#endif /* ARTERY_TRAJEKTORIAPPVERIND_H_ */

```